Programme: B.Tech
BCSE497J - Project I

Temporal Legal Drift in the Indian Code: Materiality Classification of Legislative Amendments and Evaluation of LLM-Derived Compliance Assertions

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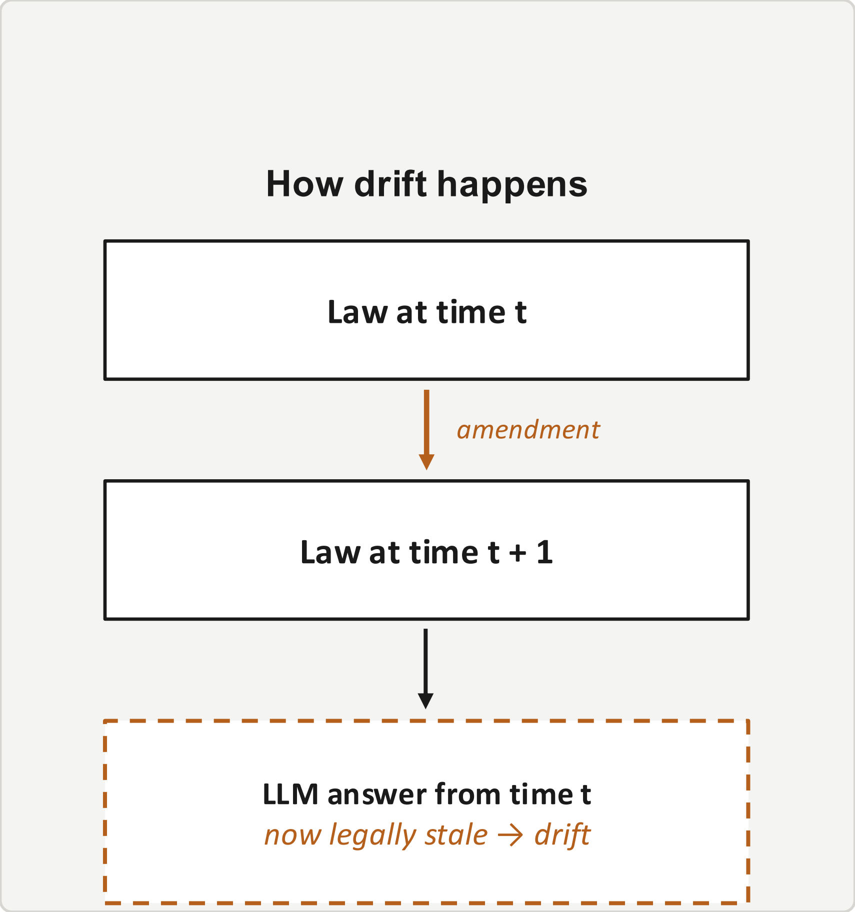
Motivation

- Indian statutes are amended constantly — provisions are inserted, substituted, omitted or repealed, changing rights, duties, thresholds and penalties.
- An LLM can give a fluent legal answer that is quietly outdated — a form of “temporal legal drift” once the underlying law moves.
- In compliance, such drift causes missed obligations, wrong risk scores and flawed reporting decisions.
- India Code offers current statutory text, but not a machine-readable amendment history or materiality record.
- Stakes are high: compliance outcomes affect regulatory exposure and legal liability.



Problem Statement

India has no unified, version-aware electronic codification of its statutes (analogous to the U.S. e-CFR) that tracks amendments over time, judges their materiality, and lets AI systems reason about whether a legal or compliance answer is still valid.

Concretely, existing systems fail to:

- Reconstruct the pre- and post-amendment version of a provision.
- Decide whether an amendment materially changes obligations or compliance outcomes.
- Detect when an LLM’s earlier compliance assertion has become legally incorrect.
- Explain, with statutory evidence, why a change matters.

Abstract

01

Goal

Build an India Code–based temporal legal benchmark that classifies amendment materiality and measures temporal drift in LLM compliance reasoning.

02

Method

Construct a version-aware corpus (India Code + amendment Acts + Gazette), define a materiality taxonomy, then evaluate LLMs on pre- vs post-amendment contexts with explainable justifications.

03

Impact

More reliable, auditable and transparent legal AI — open datasets, a drift benchmark, and a foundation for RegTech compliance monitoring in India.

Literature Survey

Paper	Objectives	Methodology	Pros/Cons	Findings
Chalkidis et al. (2022) — LexGLUE	Standardise evaluation of legal-language understanding.	Multi-task English legal benchmark; compares general and legal-domain models.	**Pros:** Standardised benchmark; domain models perform well. **Cons:** Non-Indian; static; no amendment/version analysis.	Legal-domain models outperform general models.
CUAD Hendrycks et al. (2021)	Support automated contract review and clause identification.	Expert-annotated dataset with 13,000+ legal annotations.	**Pros:** High-quality expert labels. **Cons:** Contract-focused; no temporal dimension.	Expert annotations improve legal relevance and accuracy.
XAI & Law Richmond et al. (2024)	Examine explainability in legal AI.	Evidential survey of XAI approaches in law.	**Pros:** Promotes transparency and accountability. **Cons:** No framework for amendment materiality or compliance impact.	Explainability is essential for accountable legal AI.
IL-TUR Joshi et al. (2024)	Benchmark multilingual Indian legal NLP.	Covers English, Hindi and 9 languages across NER, retrieval, prediction and statute ID.	**Pros:** Broad Indian and multilingual coverage. **Cons:** Static data; no amendments, versioning or temporal drift.	Provides extensive jurisdiction-specific Indian legal evaluation.

Paper	Objectives	Methodology	Pros/Cons	Findings
Are LLMs Court-Ready? Juvekar et al. (2025)	Assess frontier LLMs' competence in Indian law.	Indian legal exams and lawyer-graded Advocate-on-Record responses.	**Pros:** Practical, expert-evaluated assessment. **Cons:** Tests competence, not statutory evolution.	Strong on objective questions; weaker than humans in long-form reasoning.
LegalBench Guha et al. (2023)	Evaluate LLMs across diverse legal-reasoning tasks.	Collaborative benchmark containing 162 legal tasks.	**Pros:** Broad reasoning coverage. **Cons:** US-centric and static; no Indian or temporal focus.	Performance varies significantly by reasoning type.
LexTime Barale et al. (2025)	Evaluate temporal ordering in legal texts.	512 annotated event-pair instances from US federal complaints.	**Pros:** Legal-specific temporal benchmark. **Cons:** US-focused; examines event order, not legislative amendments or statute versions.	LLMs perform reasonably well, but complex legal language and nested clauses remain challenging.
TRAM Wang & Zhao (2024)	Benchmark LLM temporal-reasoning ability.	Combines 10 datasets covering event order, duration, frequency and temporal arithmetic.	**+** Broad and systematic temporal evaluation. **–** General-domain; not specific to law, legislation or amendments.	Even the best-performing LLM remains substantially below human performance.

Gap Analysis

G1	No temporal representation of Indian legislation — benchmarks use fixed texts, not amendment evolution.
G2	No materiality taxonomy — legally significant amendments are not separated from editorial ones.
G3	Temporal legal drift is unmeasured — validity of past assertions after amendments is untested.
G4	Legal change and compliance reasoning are treated as separate tasks, not integrated.
G5	Insufficient explainability for materiality, and little agentic monitoring of legislative change.

Project Objectives

1. Construct a version-aware temporal corpus of Indian legislative amendments from India Code, amendment Acts and the Gazette.

2. Develop a materiality taxonomy — High / Medium / Low / None across 10 legal dimensions (obligation, scope, threshold, liability, procedure.

3. Build a temporal legal-drift benchmark evaluating LLM compliance answers on pre- vs post-amendment versions.

4. Develop an Explainable-AI framework producing evidence-grounded materiality justifications (primary novelty).

5. Prototype an optional agentic monitor with human-in-the-loop escalation (secondary extension).

SDG Goals

SDG 9

Industry, Innovation & Infrastructure

- Builds digital legal infrastructure for amendment monitoring.
- Delivers an open temporal dataset and drift benchmark.
- Advances Legal AI, RegTech and LegalTech innovation.

SDG 16

Peace, Justice & Strong Institutions

- Improves access to reliable, current legal information.
- Produces transparent, auditable AI outputs.
- Supports accountable, contestable compliance decisions.

TRL Assessment

TRL 1–9 measures maturity, from basic principles (1) to a system proven in operation (9).

1	2	3	4	5	6	7	8	9

basic principles

proven in operation

Current **TRL 2**

Concept formulated: problem defined, literature reviewed, corpus and taxonomy designed on paper.

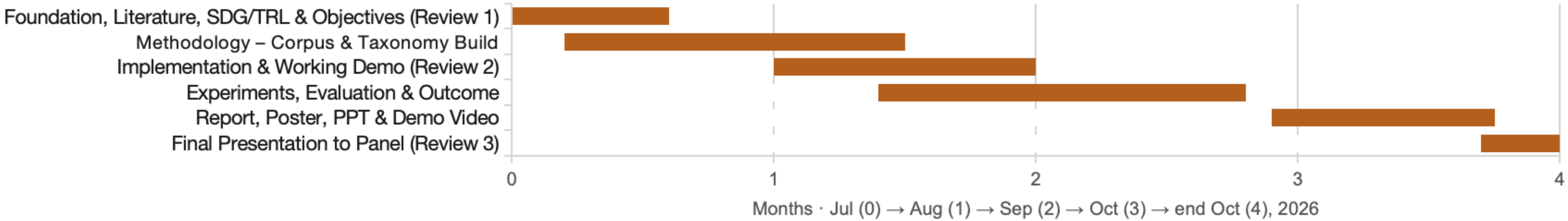
Target **TRL 4**

Validated in lab: build the corpus, run the materiality classifier and drift benchmark on India Code samples, validate against expert annotations.

Outcomes & Patents

Outcome / Deliverable	Prototype / Experiment	Timeline	Resources Required
Temporal India Code corpus + materiality benchmark	Version-aware corpus build + expert annotation study	Jul – Aug 2026 (by Review 2, 12 Aug)	India Code / Gazette access, legal annotators, storage
Explainable materiality classifier ⚑ **Patentable** *"Explainable temporal compliance architecture"*	Transformer / LLM classifier + XAI justification module	Aug – Sep 2026 (Outcome ≤ 23 Sep)	GPU compute, LLM API, PyTorch / HuggingFace
Agentic compliance monitor (secondary extension)	Human-in-the-loop legislative-monitoring prototype	Sep – Oct 2026 (by Review 3)	Orchestration framework, expert review panel

Indicative timeline (Gantt)



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